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UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2024 INVITATIONAL A

Biology		Chemistry		Physics	
B01.	B	C01.	E	P01.	C
B02.	B	C02.	C	P02.	C
B03.	E	C03.	B	P03.	E
B04.	C	C04.	A	P04.	A
B05.	D	C05.	C	P05.	D
B06.	A	C06.	B	P06.	A
B07.	D	C07.	A	P07.	B
B08.	E	C08.	E	P08.	C
B09.	D	C09.	E	P09.	C
B10.	A	C10.	B	P10.	E
B11.	E	C11.	A	P11.	D
B12.	B	C12.	D	P12.	B
B13.	A	C13.	E	P13.	E
B14.	D	C14.	D	P14.	B
B15.	C	C15.	D	P15.	C
B16.	C	C16.	B	P16.	D
B17.	E	C17.	D	P17.	D
B18.	C	C18.	E	P18.	A
B19.	A	C19.	C	P19.	B
B20.	B	C20.	A	P20.	A

CHEMISTRY SOLUTIONS – UIL INVITATIONAL A 2024

C01. (E) 1 mole = 6.022×10^{23} atoms. $(1/8)(6.022 \times 10^{23}) = 7.53 \times 10^{22}$ atoms

C02. (C) A balanced chemical equation works for molecules and for moles, and if all the reactants and products are in the gas phase, it works for liters of gas as well. But it never works for grams – in answer C there are three grams of reactants but only two grams of product.

C03. (B) A selenium atom has atomic number 34 so it has 34 protons and 34 neutrons, so a Se^{2-} ion has two extra electrons, or 36.

C04. (A) Ionic compounds are usually formed when a metal is bonded to a nonmetal.

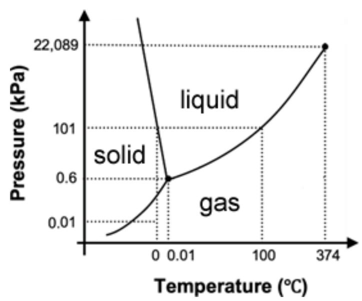
C05. (C) $P_1V_1 = P_2V_2$, so doubling the pressure reduces the volume by half.

C06. (B) $CHCl_3$ is a polar molecule, so it will have dipole-dipole and dispersion forces.

C07. (A) $\Delta U = q + w$. Changes in internal energy are equal to the heat and work associated with the change.

C08. (E). The percent composition of NaCl is: Sodium (22.99/(22.99 + 35.45) × 100 = 39.33% sodium and 60.66% chlorine, so in a 100 gram sample there would be 39.3 g of Na and 60.7 g of Cl.

C09. (E)



C10. (B) $K_P = 16 = P^2_{AB} / (P_{A2})(P_{B2})$ therefore $P^2_{AB} = 16 \times (0.5)(0.5) = 4$, so $P_{AB} = 2$ atm. The partial pressure of a gas in a mixture is the same as the pressure of that gas.

C11. (A) pH = $-\log[H^+] = -\log(3.46 \times 10^{-4}) = 3.46$. (Yes, I chose the numbers like that on purpose.)

C12. (D) $K_{sp} = [Ni^{2+}][OH^-]^2 = [x][2x]^2 = 4x^3$. $x = (2.8 \times 10^{-16}/4)^{1/3} = 4.2 \times 10^{-6} = [Ni^{2+}]$.

C13. (E) In an oxidation half-reaction electrons appear as a product.

C14. (D) If rate = $k[A]^2$ and you triple [A], the reaction will go $[3]^2 = 9$ times faster.

C15. (D) The orbital shown is a *p* orbital, so $\ell = 1$. That leaves D and E as possible answers, but E has $m_\ell = 2$, which is not an allowed value for m_ℓ when $\ell = 1$.

C16. (B) $\nu\lambda = c$ and $E = h\nu$, so $E = hc/\lambda$.
$E = (6.626 \times 10^{-34}\ J{\cdot}s)(3.00 \times 10^8\ m/sec)/(610 \times 10^{-9}\ m) = 3.26 \times 10^{-19}$ J

C17. (D) $PV = nRT$, so $n = PV/RT$. P = 1 atm, V = 10.0 L, T = 273K, R = 0.08206 L·atm/mol·K
n = (1)(10.0)/(0.08206)(273) = 0.4464 moles F_2. (Alternatively, n = 10.0/22.4 = 0.4464 moles F_2)
0.4464 moles $F_2 \times 6.022 \times 10^{23}$ molecules/mol = 2.688×10^{23} F_2 molecules
2.688×10^{23} F_2 molecules × 2 F atoms/1 F_2 molecule = 5.38×10^{23} F atoms.

C18. (E) For an ideal gas it is always true that $PV=nRT$. Therefore if n=1, it is always true that $PV=RT$, and PV/RT always equals 1, no matter what the pressure, volume, or temperature are.

C19. (C)

C20. (A) % yield = (actual/theoretical)×100%

PHYSICS SOLUTIONS – UIL INVITATIONAL A 2024

P01. (C) page 55: "…had decided to withhold from this magnum opus any mention of his ideas concerning light; he would not publish those until the bully Hooke had died – which would not happen until 1704…"

P02. (C) page 54: "Halley had been overjoyed to hear of Newton's equation of gravity; with it, he finally had been able to make sense of cometary behavior."

P03. (E) page 60: "Theirs was a dreadfully complicated task, because it required applying Newton's equation to three objects simultaneously – earth, moon, and spaceship – not just two. It was what scientists referred to as a three-body problem.", "…predicting the net effect of three objects pulling on one another, was impossible to compute exactly….the best one could hope to do was approximate an answer…"

P04. (A) Most moons in the solar system have little or no atmosphere at all. However, Titan, the largest moon on Saturn, has a thick atmosphere that is 95% nitrogen and 5% methane. The surface of Titan is shrouded by the hazy atmosphere and cannot be seen with visible light. Titan is the only moon in the solar system with a thick atmosphere.

P05. (D) The cat moves 18.0m in a time of 4.50sec. This means that the speed of the cat is $v = \frac{d}{t} = \frac{18.0}{4.50} = 4.00$m/s. Now we need to convert this speed into miles per hour: $4.00\frac{m}{s} * 60\frac{s}{min} * 60\frac{min}{hr} * \frac{1}{1609}\frac{miles}{m} = 8.95$mph.

P06. (A) We don't know how long the cat took to reach its final speed, so we will use the kinematic equation that doesn't involve time: $v_f^2 = v_i^2 + 2a\left(x_f - x_i\right)$. The initial velocity is zero, and we are given the acceleration and the distance, so: $v_f^2 = 0^2 + 2(3.50)(2.60) = 18.2 \;\; \rightarrow \;\; v_f = 4.27$m/s.

P07. (B) First, we need to construct a free-body force diagram. There is no friction, so the only forces acting on the bag are gravity (mg, downward) and the normal force (F_N, up and left, perpendicular to the inclined plane). As is usual for problems involving inclined planes, we tilt our coordinate system so that the x-axis is parallel to the plane, and the y-axis is perpendicular to the plane. In the tilted coordinate system, the normal force acts in the positive y-direction, but the gravitational force must be broken into components. The component $mg\sin\theta$ is directed down and left, parallel to the plane, in the positive x-direction; and the component $mg\cos\theta$ is directed down and right, perpendicular to the plane, in the negative y-direction. The motion of the bag of food is entirely in the x-direction, so we need only consider that component. Utilizing Newton's second law, we obtain: $\sum F_x = mg\sin\theta = ma_x$. Thus, the acceleration in the positive x-direction (down the incline) is $a_x = g\sin\theta = (9.80)\sin(28.0) = 4.60$m/s^2.

P08. (C) Work is defined as force multiplied by distance multiplied by the cosine of the angle between the force and the displacement, $W = Fd\cos\theta$. In this problem, to find the work done by friction, we first need the magnitude of the frictional force. This requires us to draw a force diagram. There are three forces acting on the toy: gravity (mg, downward, in the negative y-direction), the normal force (F_N, upward, in the positive y-direction), and the frictional force (F_f, horizontal, opposite of the motion, in the negative x-direction). There is no motion in the y-direction, so the forces in the y-direction (gravity and the normal force) must sum to zero:
$\sum F_y = F_N - mg = 0 \;\; \rightarrow \;\; F_N = mg = (0.220)(9.80) = 2.16$N. Now that we have the normal force, we can find the frictional force: $F_f = \mu F_N = (0.320)(2.16) = 0.690$N.
Finally, we can calculate the work done by friction. Because the frictional force and the motion are both horizontal, we'll use $\theta = 0°$. Thus, the work can be calculated:
$W = F_f d\cos\theta = (0.690N)(4.50m)\cos 0° = 3.10$J. Note: Technically, the displacement and frictional force are pointed in opposite directions ($\theta = 180°$), so this work would be negative – which means that the toy loses energy.

P09. (C) In order to solve this problem, we must consider the torques acting on the beam. Torque is defined as $\tau = Fr\sin\theta$, where F is a force acting on the beam, r is the distance from the pivot point to the location of the force, and θ is the angle between the force direction and the beam. For a situation like ours, the pivot point is obvious: it is the fulcrum location. We have two forces to consider – the weight of the metal beam itself, and the weight of the hanging mass. The weight of the metal beam, W, has a torque arm (the distance from the fulcrum to the force) of $r_w = 1.40$m, while the hanging mass has a torque arm of $r_h = 3.25 - 1.40 = 1.85$m. Both forces are directed vertically downward, so the angle between the force and the beam is 90° for both forces. The weight of the beam produces a torque that would cause a clockwise rotation, so it is considered negative. The hanging mass produces a torque that would cause a counterclockwise rotation, so it is considered positive. Finally, because the system is balanced, the torques must sum to zero. Mathematically: $\sum\tau = \tau_h - \tau_w = 0$. Using the formula for torque gives: $F_h r_h \sin\theta_h - W r_w \sin\theta_w = 0 \;\rightarrow\; m_h g(1.85)\sin 90° - m_w g(1.40)\sin 90° = 0$. This leads to $(25.0)(9.80)(1.85)(1) - m_w(9.80)(1.40)(1) = 0 \;\rightarrow\; 453 = 13.72 m_w$. So, the mass of the beam is $m_w = \frac{453}{13.72} = 33.0$kg.

P10. (E) First, we find the period, T, of a single oscillation of the pendulum by using the time given for ten oscillations: $T = \frac{t}{10} = \frac{28.0}{10} = 2.80$sec. Now we go to the equation relating the period of oscillation to the length of the pendulum: $T = 2\pi\sqrt{\frac{L}{g}} \;\rightarrow\; 2.80 = 2\pi\sqrt{\frac{L}{9.80}} \;\rightarrow\; \frac{2.80}{2\pi} = 0.4456 = \sqrt{\frac{L}{9.80}}$.
Thus, $L = 9.80(0.4456)^2 = 1.95$m.

P11. (D) The power lost as electromagnetic radiation by a hot object is given by $P = \sigma A e(T^4 - T_0^4)$. Here, σ is the Stephan-Boltzmann constant, A is the surface area of the object, e is the emissivity of the object, T is the temperature of the object in Kelvin, and T_0 is the ambient (room) temperature around the object (also in Kelvin). Our object has a temperature of $T = 750 + 273 = 1023$K, while the room can be assumed to be about $T_0 = 20 + 273 = 293$K. Our object is a cube, so it has six sides, and each side has a surface area of $A_{side} = (8.00cm)^2 = (0.0800m)^2 = 0.00640$m^2. So, the total surface area is $A = 6A_{side} = 6(0.00640) = 0.0384$m^2. Thus, the total power output as electromagnetic radiation from the heated cube is
$P = (5.67\times10^{-8})(0.0384)(0.750)[(1023)^4 - (293)^4] = 1776W \approx 1780W$. Note: the ambient (room) temperature doesn't make much difference in this problem since the cube is so hot. You will still get the right answer even if you ignore the room temperature contribution.

P12. (B) First, we combine the resistors to find a single equivalent resistance. Since the resistors are in series, we combine them by adding them: $R_{total} = R_1 + R_2 = 220 + 470 = 690\Omega$. This is the total circuit resistance. Now we can use Ohm's Law to find the total current produced by the battery: $I_{total} = \frac{V_{total}}{R_{total}} = \frac{24.0}{690} = 0.03478$A. Since everything in this circuit is in series, this same current flows from the battery and through each of the two resistors. Thus, by series rules: $I_1 = I_2 = I_{total} = 0.03478$A. Now we return to Ohm's Law to find the voltage drop across the 470Ω resistor. $V_2 = I_2 R_2 = (0.03478)(470) = 16.3$V.

P13. (E) Since there is only a single charge, we can find the magnitude of the electric field simply by using $|E| = \frac{kQ}{r^2}$. The distance from the charge Q to the point P is found from the coordinate system:
$r = 80.0 - 20.0 = 60.0cm = 0.600m$. Now, the magnitude of the electric field is
$|E| = \frac{(8.99\times10^9)(250\times10^{-9})}{(0.600)^2} = 6240$N/C.

P14. (B) We find the magnitude of the force on the current-carrying wire by using $|F| = ILB\sin\theta$, where I is the current in the wire, L is the length of the wire in the magnetic field, B is the magnitude of the magnetic field, and θ is the angle between the field direction and the current direction. Fortunately, the field direction and the current direction are perpendicular, so $\theta = 90°$. Putting this all together gives $|F| = (14.0)(0.240)(0.650)\sin 90 = 2.18\text{N}$. Using the right-hand rule, we have the current (index finger) pointed in the positive x-direction, and the magnetic field (middle finger) pointed in the positive z-direction. This gives a force (thumb) pointed in the negative y-direction (down). Thus, the correct answer is that $F = 2.18N$ down (negative y-direction).

P15. (C) Since the laser beam is initially vertically polarized, we only need to use Malus' Law once to calculate the intensity of the beam after it passes through the tilted polarizer. Malus' Law is $I = I_0(\cos\theta)^2$ where θ is the angle between the initial polarization of the beam and the polarizer. In this case that is given as $\theta = 32.0°$. So, the final intensity is $I = (500)(\cos 32)^2 = 360\text{W/m}^2$.

P16. (D) First, we need to find the location of the image of the butterfly by using $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$. Putting in the object location $p = 29.0\text{cm}$ and the focal length $f = -18.0\text{cm}$, we get an image location of $\frac{1}{(29.0)} + \frac{1}{q} = \frac{1}{(-18.0)} \rightarrow q = -11.1\text{cm}$. From this, we can determine the magnification of the image: $M = -\frac{q}{p} = -\frac{-11.1}{29.0} = 0.383$.

P17. (D) The spectral line wavelength allows us to determine the energy of the photons emitted during the transition: $E = \frac{1240eVnm}{\lambda} = \frac{1240}{713} = 1.74\text{eV}$. This energy must be equal to the difference between two of the energy levels of the element. From the energy level diagram, we can see that the $2^3P \rightarrow 1^1S$ difference exactly equals the photon energy we calculated. Thus, the strong spectral line is caused by the $2^3P \rightarrow 1^1S$ transition. Note: the other energy differences are 2.76eV, 1.44eV, 1.32eV, 1.02eV, and 0.30eV; so, none of the other possible transitions give the correct photon energy.

P18. (A) When a particle and its antiparticle interact, it will end with the particles annihilating one another and producing gamma rays. It does not matter if they are hadrons or leptons, massive or light, charged or uncharged: matter-antimatter interactions always end the same way. The products of this interaction, gamma rays, represent pure electromagnetic energy. This result also tells us which force is responsible: The Electromagnetic Force.

P19. (B) We are given a plot of velocity versus time. Acceleration is represented by a slope on this plot. Instantaneous acceleration is represented by the slope of the curve at a particular place on the plot – in our case at a time of five seconds. The point of interest lies in the middle section of the plot, and the line segment in this section has a slope of $slope = a = \frac{6.0m/s - 2.0m/s}{8.0s - 4.0s} = \frac{4.0m/s}{4.0s} = 1.0\text{m/s}^2$. This is the acceleration of the bird at $t = 5.0$seconds.

P20. (A) Electric potential is given by $V = \frac{kQ}{r}$. Since the plot is of V versus $\frac{1}{r}$, we can see from the equation that the slope is equal to kQ. To calculate the slope, I'll use the points (1.5m^{-1}, 375V) and (0.5m^{-1}, 125V). Then: $slope = \frac{375V - 125V}{1.5m^{-1} - 0.5m^{-1}} = \frac{250Vm}{1.0} = 250Vm$. Relating the slope to our equation and plugging in the Coulomb constant gives: $kQ = 250Vm = (8.99 \times 10^9)Q \rightarrow$ $Q = 2.8 \times 10^{-8}C = 28 \times 10^{-9}\text{C}$. Thus, the magnitude of the point charge is $Q = 28\text{nC}$.